

Aly Shahbaz Aftab Khan

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EDUCATION

University of Nottingham

09/2022-07/2025

- Computer Science with Artificial Intelligence BSc (Hons) | **First Class Honors**. GPA (Converted): **4.0**
- *Relevant Modules* – Machine Learning (**81%**); Computer Vision (**80%**)
- *Additional Training*: Software Engineering – J.P. Morgan Chase & Co., Software Engineering – Goldman Sachs, Software Engineering - Citi (All completed via Forage)
- *Other Academic Achievements*: 40/45 in IBDP (A-Level Equivalent), November 2021 – Highest Score in Pakistan. Ranked 3rd nationally in the Caribou Mathematics Competition.

WORK EXPERIENCE

Software Engineering Intern at Oii.ai

05/2025-08/2025

- Designed and implemented a secure system leveraging locally run LLMs with function calling to query a GraphQL database, ensuring 100% of client data remained abstracted from the language model.
- Eliminated hallucinations and enforced deterministic outputs by replacing model reasoning with chained functions.
- Enabled inter-function calling to minimise repetition and reduce codebase complexity.
- Produced a comprehensive testing suite to validate function execution, data integrity, and edge-case handling, improving system reliability and reducing user error by 15%.

Machine Learning Intern at Nestlé

08/2024-09/2024

- Worked alongside a 50-person Supply Chain Management team to design and implement Machine Learning projects in Python.
- Created a sentiment analysis tool that tracks product sentiment over 3 different social media platforms and outputs a product popularity score.
- Engineered a machine learning model to forecast product revenue based on comparable products, achieving a 7% error rate in back-testing and enabling more accurate demand planning.

Artificial Intelligence Development Intern at Tintash

07/2024-08/2024

- Built a game-development-specific LLM plugin for Unity using C# as part of a 3-person team.
- Delivered proof of concept by implementing a code generator that created gameObjects and generated scripts automatically using the ChatGPT API, reducing the time to code a small game by almost 30%.
- Completed a standalone project, produced an LLM-powered multi-agent system using Python and Langchain, where each of the 5 agents communicated to guess a user-set object.

Quantitative Developer/Software Engineer at Nottingham Economics and Finance Society

10/2023-04/2024

- 1/25 selected from 92 applicants, engineered a FOREX GBP/USD trading strategy that returned a simulated average profit of 13% over the past five years.
- As a quantitative developer, helped design the trading strategy by analysing 10 years' worth of data to identify historical trends.
- As a software engineer, built and continuously updated the trading algorithm in Python. Also implemented back-testing functions that simulated the strategy's performance.

Software Engineering Team Leader at University of Nottingham & Beckford Consulting

10/2023-04/2024

- Led a team of 8 student software engineers in developing software for a partner company of the university.
- Roles included communicating with all stakeholders, planning the software development life cycle, holding the team to the principles of AGILE development, and presenting the four deliverables on time.
- Produced a mobile application and an SQL database for 300 students and instructors using React Native.

RELEVANT PROJECTS

[Handwritten Character Recognition CNN](#)

- Developed a CNN that recognises handwritten characters using TensorFlow, which returns a balanced accuracy of 99% in test sets.
- Preprocessed the dataset of 120,000 handwritten characters and designed a model that uses batch normalisation and dropout layers to optimise model performance.
- Added functionality allowing a user to draw any character and return the model's prediction after automatic preprocessing.

SKILLS

Languages: Python, Java, JavaScript, C++, C#

ML/AI Tools: TensorFlow, Scikit-learn, Langchain, NLTK, NLP

Web & App Dev: React, React Native, TailwindCSS, SQL

Other Tools: Git, MySQL, AWS, LINUX

Development Practices: AGILE, TDD, SDLC